

Sean Luka Girgis

Lead Performance / Test Enablement Engineer | Python, NFRs, Test Strategy & Quality Metrics

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Senior performance engineering, test enablement, and production quality professional with 20+ years of enterprise technology experience across performance testing, test strategy, NFR evaluation, defect triage, monitoring/APM, Python and scripting, testing/reporting artifacts, stakeholder reporting, mentoring, and cross-functional delivery. Strong background analyzing application behavior, identifying bottlenecks, validating production readiness, documenting test findings, building operational dashboards, and communicating quality and performance risks to engineering and leadership teams. Best aligned to testing enablement roles requiring performance engineering, test strategy, defect management, test metrics, CI/CD awareness, Agile/Waterfall collaboration, mentoring, and Python/scripting; Playwright, JUnit, TestNG, TestComplete, GitLab CI, NeoLoad, and formal automation-suite ownership are positioned carefully as adjacent or working-knowledge areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported performance, capacity, monitoring, and production-quality workflows across large-scale banking infrastructure using BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and telemetry pipelines.
- Built Python/Pandas/PySpark and SQL workflows that transformed telemetry from 6,000+ infrastructure endpoints into validated dashboards, capacity forecasts, quality checks, and executive-ready reporting.
- Developed validation and regression-style checks to compare current outputs against prior trusted runs, identify abnormal metric movement, and prevent inaccurate reporting from reaching stakeholders.
- Investigated production quality issues by tracing source feeds, transformation logic, SQL behavior, metric behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Created KPI-based reporting and operational metrics that helped teams understand performance trends, bottleneck risks, utilization patterns, and service-health indicators.
- Partnered with infrastructure, application, analytics, engineering, and business teams to prioritize issues, explain findings, coordinate fixes, and improve operational reliability.
- Documented test/validation logic, metric definitions, data flow notes, support procedures, runbooks, and troubleshooting steps to improve repeatability and team handoff.
- Used Python, SQL, Unix/Linux-style automation patterns, Git concepts, and CI/CD-aware delivery discipline to support repeatable validation, reporting, and production troubleshooting workflows.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed performance analysis and monitoring for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.
- Led the FAST project, mining real-user and synthetic monitoring metrics to identify performance improvement opportunities adopted by engineering teams.

- Correlated load-test behavior with APM telemetry to identify bottlenecks, user-impacting symptoms, and improvement areas before production impact.
- Created dashboards and reporting views that converted monitoring and test signals into actionable performance, reliability, and stakeholder insights.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Defined and promoted monitoring standards, dashboards, alert rules, SLA thresholds, test/production visibility practices, and operational reporting artifacts across technical teams.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable quality checks.
- Provided mentoring, onboarding support, troubleshooting guidance, and documentation to improve team adoption of monitoring and performance-analysis practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, NFR-style evaluation, and release-readiness analysis.
- Analyzed J2EE application behavior under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented test results, expected behavior, defects, risks, performance metrics, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and performance-focused cutover support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported test planning, phased validation, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, and messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, and troubleshooting for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

FAST Performance Analytics Project

Mined Dynatrace and Gomez Synthetic Monitoring data to identify optimization opportunities, improve transaction visibility, and communicate actionable performance findings to engineering teams.

Technologies: Dynatrace AppMon, Gomez Synthetic Monitoring, Performance Analytics, Dashboards, APM

Production Data Quality and Regression Checks

Built validation and comparison workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, and prior-run comparisons.

Technologies: Python, SQL, Pandas, Validation, Regression Checks, Runbooks

Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.

Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation